Quiz–1 — Solution Key

Foundations of Fluid Mechanics, AM2530

Instructor: Dr. Anubhab Roy

02/09/2026 50 minutes Maximum marks: 30

Award full credit for any correct alternative route. Carry-forward errors are penalised once only. Answers are quoted to three significant figures; accept ±1% for rounding and $g = 9.8$ or 9.81 m/s^2.

Q1. (6 marks)

(a) **Identification.** (**4 marks**)

(i) **Streakline.** The chimney emits continuously from one fixed point, so a single instantaneous photograph shows all the particles that have left that point at earlier times, strung into one curve.

(ii) **Streamlines.** The arrows are the velocity field at one instant and the traces are constructed everywhere tangent to that instantaneous field.

(iii) **Pathlines.** Each burning particle is one material element followed over a finite exposure, so its track is its own trajectory.

(iv) **Timeline.** A set of neighbouring fluid elements is marked *simultaneously* along a line right across the channel; the later curves show how the flow has deformed that one material line, carrying it fastest in the core and holding it back at the no-slip walls.

Marking: 1 mark each. Award the mark for the correct name provided the justification is not actively wrong. The discriminating pair is (i) versus (iii): continuous release from one point seen at one instant is a streakline; one particle followed in time is a pathline. If a student swaps these two, deduct both.

(b) **Coincidence in steady flow.** (**2 marks**)

Streamlines, pathlines and streaklines coincide if the flow is steady. If the velocity field at every fixed point is frozen in time, a particle released at a point always sees the same direction field, so the curve it traces out is the very streamline through its release point; and every later particle released from that same point retraces the identical curve, which makes the streakline that curve too.

The **timeline** does not join them. It is a material line laid *across* the flow at one instant; it is not built from the velocity direction at all and is in general nowhere tangent to $\vec{V}$. The flow merely stretches and tilts it, so there is nothing for it to coincide with — and this remains true whether the flow is steady or not.

Marking: 1 mark for naming the correct three with any workable reason; 1 mark for a correct statement of why the timeline is different in kind (a material line, not a curve built from the velocity direction). "Because a timeline depends on time" is not sufficient — so do streaklines.

Q2. (12 marks)

(a) **Air pressure and the equivalent free surface.** (**5 marks**)

The air is at rest and its density negligible, so the pressure at the air–water interface is p_A. Stepping through the manometer from the air space to the open limb, pressure increasing downwards within each continuous constant-density column,

$$p_A + \rho_w g\,(1.2 + 0.30) - \rho_{Hg}\,g\,(0.50) = p_{\text{atm}},$$

$$p_A = (13600)(9.81)(0.50) - (1000)(9.81)(1.50) = 66{,}708 - 14{,}715 = \boxed{52.0 \text{ kPa (gauge)}}.$$

The equivalent free surface stands

$$h_e = \frac{p_A}{\rho_w g} = \frac{51{,}993}{9810} = \boxed{5.30 \text{ m}}$$

above the actual interface. Every hydrostatic calculation in this tank may then be done as if the tank were open, with all depths measured from a surface 5.30 m higher than the real one.

Marking: 1 mark for a correctly signed manometer walk; 1 mark for including the full 1.5 m of water (the 1.2 m inside the tank plus the 0.30 m in the tubing) — using the 0.30 m alone is the standard slip and costs this mark; 1 mark for the arithmetic giving $p_A \approx 52$ kPa gauge; 1 mark for recognising $h_e = p_A/\rho_w g$; 1 mark for $h_e = 5.30$ m together with a statement of what it buys — that all depths may now be measured from the raised surface.

(b) **Components on the fillet (per metre width). (7 marks)**

Horizontal. The horizontal force on a curved surface equals that on its vertical projection: a 1.5 m tall strip spanning depths 4.0–5.5 m below the real interface, i.e. 9.30–10.80 m below the equivalent surface, with centroid $h_c = 10.05$ m:

$$F_H = \rho_w g\, h_c\, (R \times 1) = (1000)(9.81)(10.05)(1.5) = \boxed{147.9 \text{ kN}},$$

acting **horizontally, from the water towards the wall**. Equivalently $F_H = p_A R + \rho_w g\,(5.5^2 - 4.0^2)/2 = 77{,}990 + 69{,}896$ N.

Vertical. The vertical force is $\int p\, dA$ over the *horizontal* projection of the fillet, which is a strip of width R. On that projection the pressure is $p = p_A + \rho_w g\,\eta$, so the trapped air contributes a term of its own:

$$F_V = \underbrace{p_A\, R}_{\text{air on the projection}} + \underbrace{\rho_w g \int_0^R \eta_{\text{arc}}\, d\xi}_{\text{weight of the water above the fillet}},$$

$$\int_0^R \eta_{\text{arc}}\, d\xi = (4.0)R + \frac{\pi R^2}{4} = 6.000 + 1.767 = 7.767 \text{ m}^2.$$

$$F_V = (51{,}993)(1.5) + (1000)(9.81)(7.767) = 77{,}990 + 76{,}196 = \boxed{154.2 \text{ kN}},$$

acting **vertically downwards**, since the water lies above the fillet. Note that the air term is the larger of the two, 51% of the total: it is *not* a small correction, and dropping it would put F_V out by a factor of two.

Equivalently, in one step with the equivalent free surface, $p_A R$ is just the weight of the extra column of height h_e standing on the same projection:

$$F_V = \rho_w g \left[(4.0)R + \frac{\pi R^2}{4} + h_e R\right] = (1000)(9.81)\big[6.000 + 1.767 + 7.950\big] = 154.2 \text{ kN}.$$

Marking: Horizontal, 3 marks: *1 for taking F_H on the vertical projection of the fillet; 1 for the correct centroidal depth on the equivalent-surface datum, or equivalently for an explicit $p_A R$ term; 1 for $F_H = 147.9$ kN.* Vertical, 3 marks: *1 for the $p_A R$ term; 1 for the $\rho g \times$ (area above the surface) term with the quarter circle included; 1 for $F_V = 154.2$ kN.* Directions, 1 mark: *both stated correctly. The $p_A R$ term is the single most likely thing to be dropped — many students carry p_A into F_H, where it is visible in the pressure distribution, and then compute F_V as "the weight of the water above the surface" alone, which halves the answer. $F_V \approx 76$ kN is the signature of exactly that error. Omitting p_A from both components scores at most 2 of 7.*

Q3. (12 marks)

(a) **Momentum-deficit form of the drag.** (**5 marks**)

Take a fixed rectangular control volume of width b, face 1 far upstream where the flow is uniform at U_0, face 2 at the measured station, and the upper and lower faces at $z = \pm Y$ far enough out that $u = U_0$ there. The static pressure is uniform over the whole control surface, so the pressure forces cancel.

Mass. The wake carries less mass than the uniform stream, so fluid is expelled sideways:

$$\rho U_0 (2Y) b = \rho b \int_{-Y}^{Y} u \, dz + \dot{m}_{\text{side}} \implies \dot{m}_{\text{side}} = \rho b \int_{-Y}^{Y} (U_0 - u) \, dz \; > \; 0.$$

Momentum. That fluid leaves through the upper and lower faces, where the axial velocity is U_0, so it removes x-momentum at the rate $\dot{m}_{\text{side}} U_0$. The only force on the control volume is the reaction of the plate, $-D$, so

$$-D = \rho b \int u^2 dz + U_0 \dot{m}_{\text{side}} - \rho U_0^2 (2Y) b.$$

Substituting $\dot{m}_{\text{side}}$ and noting $\rho b U_0 \int_{-Y}^{Y} U_0 \, dz = \rho b U_0^2 (2Y)$, the last term cancels:

$$-D = \rho b \int_{-Y}^{Y} u^2 dz - \rho b U_0 \int_{-Y}^{Y} u \, dz.$$

Outside the wake the integrand $u(U_0 - u)$ vanishes identically, so nothing more is picked up as the faces are moved out, and letting $Y \to \infty$ gives

$$\boxed{D = \rho \, b \int_{-\infty}^{\infty} u \, (U_0 - u) \, dz}.$$

Marking: 1 mark for a drawn control volume with its four faces identified and the limits $z = \pm Y$ marked; 1 mark for the mass balance identifying $\dot{m}_{side}$ as the deficit flux leaving sideways; 1 mark for crediting that flux with axial velocity U_0 in the momentum balance — omitting this term is the standard error and gives the wrong answer; 1 mark for recognising that the pressure is uniform over the whole control surface so the pressure forces cancel, leaving $\sum F_x = -D$; 1 mark for the algebra and for taking $Y \to \infty$ to reach the final result. A student who never draws the control volume, however correct the algebra, loses the first mark.

(b) **Evaluating the drag.** (**4 marks**)

In the dead-air region $u = 0$, so $u(U_0 - u) = 0$ and it contributes nothing. For $|z| > L/2$ put $s = |z| - L/2$:

$$u = U_0 \left(1 - e^{-s/L}\right), \quad U_0 - u = U_0 e^{-s/L}, \quad u(U_0 - u) = U_0^2 \left(e^{-s/L} - e^{-2s/L}\right).$$

Doubling the upper half by symmetry,

$$D = 2\rho b U_0^2 \int_0^{\infty} \left(e^{-s/L} - e^{-2s/L}\right) ds = 2\rho b U_0^2 \left(L - \frac{L}{2}\right) = \boxed{\rho \, U_0^2 L \, b}.$$

With the given data,

$$D = (1.20)(12)^2 (0.40)(1.5) = \boxed{103.7 \text{ N}}, \qquad C_D = \frac{\rho U_0^2 L b}{\frac{1}{2} \rho U_0^2 L b} = \boxed{2.00}.$$

The measured drag coefficient of a two-dimensional flat plate normal to a stream is about 2, so this crude two-piece profile lands on the right answer. The agreement is less impressive than it looks: the model contains one free parameter, the decay length of the exponential, and writing it equal to the plate height L is what produces $C_D = 2$. A decay length λ would have given $D = \rho U_0^2 \lambda b$ and $C_D = 2\lambda/L$.

Marking: 1 mark for substituting the profile and recognising that the dead-air core contributes zero; 1 mark for the two exponential integrals (L and $L/2$); 1 mark for the symmetry factor of 2 and the closed form $D = \rho U_0^2 Lb$; 1 mark for $D = 103.7$ N and $C_D = 2.00$. Losing the factor of 2 from symmetry costs 1 mark only. The closing remark about the decay length is a bonus and carries no marks. Losing the factor of 2 costs 1 mark only.

(c) **Raising U_0 until the wake sheds. (1 mark)**

The term is the **unsteady, or storage, term** of the Reynolds Transport Theorem,

$$\boxed{\frac{\partial}{\partial t}\int_{CV} \rho\,\vec{V}\; d\forall},$$

the rate of change of the momentum held inside the control volume. In part (a) the flow was steady and it vanished. Once the plate sheds vortices the momentum inside the control volume rises and falls as each vortex forms and is swept away, so the term is non-zero at every instant and the drag fluctuates with it. (It would still average to zero over a whole number of shedding periods, since the control volume then returns to the state it started in — but that is not asked for here.)

Marking: 1 mark for naming the unsteady term. Accept "the storage term", "the rate of change of momentum inside the control volume", "the $\partial/\partial t$ term", or the term written out. Do not require any discussion of averaging.

(d) **Why the dead air contributes nothing. (2 marks)**

The integrand is a product of two things: the mass flux $\rho u\, dz$ that actually crosses the downstream face, and the velocity defect $(U_0 - u)$ that this fluid carries. Where $u = 0$, *no fluid crosses that part of the face at all*; it transports neither mass nor momentum, so it cannot register a deficit there, however wide it is.

The mass that would have passed through the dead-air region has not vanished: it was pushed sideways and left through the upper and lower faces, and its momentum was accounted for exactly once, in the $U_0\dot{m}_{\text{side}}$ term of part (a), which cancelled against part of the upstream flux. The drag is registered entirely in the shear layers flanking the dead air, where $0 < u < U_0$.

Nor does this make the plate size irrelevant: $D = \rho U_0^2 Lb$ is proportional to L, because the dead-air region sets the scale over which the surrounding shear layers recover, even though it contributes nothing pointwise.

Marking: 1 mark for the central point — zero velocity means zero mass flux across that part of the face, hence no momentum transport there; 1 mark for accounting for where the missing mass went, namely out through the top and bottom faces, already included in part (a); locating the drag in the flanking shear layers, or noting that D still scales with L, is creditable towards either mark. A student who writes only "because $u = 0$ makes the integrand zero" has restated the algebra and earns nothing.

Coverage against the Quiz 1 syllabus (L1–L15). Flow classification: streamlines, pathlines, streaklines and timelines — Q1, 6 marks. Fluid statics, pressure equation, gauge versus absolute pressure, manometry — Q2(a), 5 marks. Hydrostatic force on a curved surface, horizontal and vertical components — Q2(b), 7 marks. System versus control volume, Reynolds Transport Theorem, integral mass conservation and linear momentum for an inertial control volume applied to flow past a body — Q3, 12 marks. **Total: 30.** *Not examined on this paper: buoyancy, Archimedes' principle and metacentric stability; and, beyond the visual classification in Q1, the material derivative and the acceleration field.*